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Comment on “A comparison of the von Mises and Hencky equivalent strains for use in simple shear experiments”

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Onaka has shown (2010) that the Hencky equivalent strain is the appropriate quantity to represent the degree of severe plastic deformation based on simple-shear deformation. In a recent paper, Shrivastava et al. have criticized (2011) the Onaka paper and stated that the Hencky formulation does not apply to large simple-shear deformation. In this response, it is shown that this claim of Shrivastava et al. is not consistent with recent accepted knowledge on the Hencky strain.

Keywords: severe plastic deformation; Hencky strain; simple-shear deformation; equivalent strain

1. Introduction

Severe plastic deformation (SPD) is an effective method to produce ultra-fine grained materials [1]. There are several different SPD processes and some of them are based on simple-shear deformation [1–3]. To compare the degree of SPD produced by different processes, an appropriate scalar representing the degree of deformation is needed. Onaka has shown that the Hencky equivalent strain is the appropriate quantity to represent the degree of SPD [2,3]. However, in a recent paper, Shrivastava et al. [4] have criticized the Onaka paper [3] and stated that the Hencky formulation does not apply to large simple-shear deformation, because the axes of the Hencky strain and its increment do not coincide. In this response, it is shown that this claim of Shrivastava et al. is contrary to recent accepted knowledge on the Hencky strain.

2. The Hencky strain

Here is a brief outline of the Hencky strain. The deformation-gradient tensor $\mathbf{F}$ is the fundamental quantity for defining strain. $\mathbf{F}$ is decomposed via the polar decomposition theorem [5] into:

$$\mathbf{F} = \mathbf{R}\mathbf{U} = \mathbf{V}\mathbf{R}, \quad (1)$$

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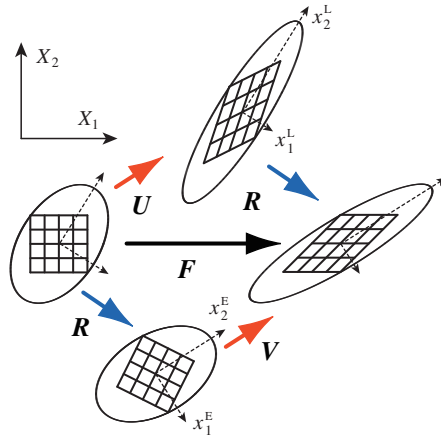


Figure 1. Polar decomposition of the deformation gradient tensor $\mathbf{F}$ for simple-shear deformation.

where $\mathbf{R}$ is the orthogonal rotation tensor, and the right-stretch tensor $\mathbf{U}$ and the left-stretch tensor $\mathbf{V}$ are symmetric tensors with positive eigenvalues. The tensor $\mathbf{R}$ represents pure rotation in $\mathbf{F}$. On the other hand, $\mathbf{U}$ and $\mathbf{V}$ represent the essential deformation in $\mathbf{F}$. Figure 1 shows the operations of $\mathbf{U}$, $\mathbf{V}$ and $\mathbf{R}$ in the case of simple-shear deformation for

$$\mathbf{F} = \begin{pmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (2)$$

where γ represents the amount of simple shear. In Figure 1, grids and ellipses around the grids show the shapes before and after the occurrence of $\mathbf{U}$, $\mathbf{R}$, $\mathbf{V}$ and $\mathbf{F}$. The Lagrangian Hencky strain $\mathbf{h}_L$ and the Eulerian Hencky strain $\mathbf{h}_E$ are defined as [6]:

$$\mathbf{h}_L = \ln \mathbf{U} \quad \text{and} \quad \mathbf{h}_E = \ln \mathbf{V}, \quad (3)$$

where $\ln \mathbf{U}$ and $\ln \mathbf{V}$ are the logarithms of $\mathbf{U}$ and $\mathbf{V}$ [7]. Using the axes X_i in Figure 1, the components of $\mathbf{h}_L$ are written as

$$\mathbf{h}_L = f(\gamma) \begin{pmatrix} -\gamma & 2 & 0 \\ 2 & \gamma & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (4)$$

where:

$$f(\gamma) = \ln \left[(\gamma/2) + \sqrt{1 + (\gamma/2)^2} \right] / \sqrt{4 + \gamma^2}. \quad (5)$$

On the other hand, using the axes of the principal strain x_i^L in Figure 1, the components of the Lagrangian Hencky strain are written as

$$\mathbf{h}_L^p = \ln \left[(\gamma/2) + \sqrt{1 + (\gamma/2)^2} \right] \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (6)$$

The components of $\mathbf{h}_E$ and $\mathbf{h}_E^p$ are obtained by changing the 1–1 and 2–2 components of $\mathbf{h}_L$ and $\mathbf{h}_L^p$, for example:

$$\mathbf{h}_E = f(\gamma) \begin{pmatrix} \gamma & 2 & 0 \\ 2 & -\gamma & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (7)$$

3. Rotation of the principal axes of the Hencky strain

The relationship between $\mathbf{h}_L$ and $\mathbf{h}_L^p$ is given by

$$\mathbf{h}_L = \mathbf{T}^T \mathbf{h}_L^p \mathbf{T}, \quad (8)$$

where $\mathbf{T}$ is the orthogonal rotation tensor written as [4]

$$\mathbf{T} = \begin{pmatrix} \sqrt{\frac{1}{2} + \frac{\gamma/2}{\sqrt{4 + \gamma^2}}} & -\sqrt{\frac{1}{2} - \frac{\gamma/2}{\sqrt{4 + \gamma^2}}} & 0 \\ \sqrt{\frac{1}{2} - \frac{\gamma/2}{\sqrt{4 + \gamma^2}}} & \sqrt{\frac{1}{2} + \frac{\gamma/2}{\sqrt{4 + \gamma^2}}} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (9)$$

and $\mathbf{T}^T$ is the transpose of $\mathbf{T}$. $\mathbf{T}$ is a function of γ and the principal axes of the Hencky strain rotate during simple-shear deformation. From Equation (7), the increment $d\mathbf{h}_L^p$ of $\mathbf{h}_L^p$ is written as

$$d\mathbf{h}_L^p = \frac{d\gamma}{\sqrt{4 + \gamma^2}} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (10)$$

To discuss the increment of the Hencky strain observed on the X_i axes, Shrivastava et al. [4] have considered the following relationship obtained by differentiating both sides of Equation (8) with respect to γ :

$$d\mathbf{h}_L = \mathbf{T}^T d\mathbf{h}_L^p \mathbf{T} + d\mathbf{T}^T \mathbf{h}_L^p \mathbf{T} + \mathbf{T}^T \mathbf{h}_L^p d\mathbf{T}. \quad (11)$$

However, as will be shown later in this paper, this relationship is the reason why Shrivastava et al. have misunderstood the appropriateness of the Hencky formalism. To reveal the problem in Equation (11), we consider the strain rate of simple-shear deformation in the next section.

4. Strain rate of simple-shear deformation

The strain $\mathbf{h}_L$ in the left-hand-side of Equation (8) is that observed on the rotating x_i^L axes and expressed on the X_i axes. When the simple-shear deformation shown in Figure 1 occurs under the rate of $d\gamma/dt = \dot{\gamma}$, from Equations (4), (5) and (8)–(11), the time derivative of $\mathbf{h}_L$, $d\mathbf{h}_L/dt = \dot{\mathbf{h}}_L$ is given by

$$\dot{\mathbf{h}}_L = \frac{\dot{\gamma}}{4 + \gamma^2} \begin{pmatrix} -\gamma - 4f(\gamma) & 2 - 2\gamma f(\gamma) & 0 \\ 2 - 2\gamma f(\gamma) & \gamma + 4f(\gamma) & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (12)$$

The above $\dot{\mathbf{h}}_L$ has the same form as the strain increment $d\mathbf{h}_L$ obtained by Shrivastava et al. [4]. However, it may be difficult to accept this complicated $\dot{\mathbf{h}}_L$ as the strain rate of simple-shear deformation expressed on the X_i coordinate system.

From a metallurgical viewpoint, the simple-shear deformation could be caused by glide motion of dislocations. If the glide motion of dislocations satisfying $\dot{\gamma} = \rho b v$ occurs as shown in Figure 2, the simple-shear deformation in Figure 1 continues and the value of γ increases. Here ρ is the dislocation density, b the magnitude of the Burgers vector and v the velocity of the dislocations. This intuition tells us that, regardless of the value of γ , the strain rate $\dot{\epsilon}$ of the simple-shear deformation should have the form:

$$\dot{\epsilon} = \begin{pmatrix} 0 & \dot{\gamma}/2 & 0 \\ \dot{\gamma}/2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (13)$$

On the other hand, from a mechanics viewpoint, the strain rate is given by the rate of deformation $\mathbf{D}$ [5,8,9]. $\mathbf{D}$ is the symmetric part of the velocity gradient $\mathbf{L}$,

$$\mathbf{D} = \frac{1}{2}(\mathbf{L} + \mathbf{L}^T), \quad (14)$$

and $\mathbf{L}$ is given by

$$\mathbf{L} = \dot{\mathbf{F}}\mathbf{F}^{-1}. \quad (15)$$

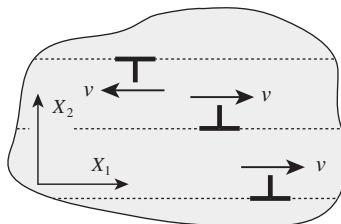


Figure 2. Simple-shear deformation caused by glide motion of dislocations satisfying $\dot{\gamma} = \rho b v$, where ρ is the dislocation density, b the magnitude of the Burgers vector and v the velocity of the dislocations.

For simple-shear deformation with $\mathbf{F}$ given by Equation (2), the rate of deformation $\mathbf{D}$ becomes:

$$\mathbf{D} = \begin{pmatrix} 0 & \dot{\gamma}/2 & 0 \\ \dot{\gamma}/2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (16)$$

which is the same as Equation (13). As a result, the rate $\dot{\mathbf{h}}_L$ given by Equation (12) could not be accepted as the strain rate of simple-shear deformation [8,9].

Although Equation (16) is the same as the strain rate given by the infinitesimal or nominal strain, $\mathbf{D}$ is essentially different from the strain rate of the infinitesimal strain. For example, $\mathbf{D}$ for rolling deformation [2,3] with

$$\mathbf{F} = \begin{pmatrix} 1 + \alpha & 0 & 0 \\ 0 & (1 + \alpha)^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (17)$$

is written as

$$\mathbf{D} = \begin{pmatrix} \dot{\alpha}(1 + \alpha)^{-1} & 0 & 0 \\ 0 & -\dot{\alpha}(1 + \alpha)^{-1} & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (18)$$

where α represents the amount of rolling. This is the strain rate of the logarithmic strain given by

$$\mathbf{h} = \begin{pmatrix} \ln(1 + \alpha) & 0 & 0 \\ 0 & -\ln(1 + \alpha) & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (19)$$

5. Applicability of the Hencky strain

Historically, the Hencky strain was used in cases where the principal strain axes do not rotate during deformation and $d\mathbf{T} = d\mathbf{T}^T = 0$ is satisfied in Equation (11). It is true that consideration of the change in the Hencky strain by Equation (11) causes problems when $d\mathbf{T}, d\mathbf{T}^T \neq 0$. Fitzgerald [10] has derived essentially the same result as Equation (11) in 1980 and described that the utility of the Hencky strain is limited to situations with fixed principal strain. In papers published in the 1980s [10–13], we find the consensus on the inapplicability of the Hencky strain to cases where the principal strain axes rotate during deformation. This consensus is the same as the claim of Shrivastava et al. [4] that the Hencky formulation does not apply to large simple shear deformation because the axes of the Hencky strain rotate during deformation.

However, even if $\dot{\mathbf{h}}_L$ as given by Equation (12) is not the strain rate, this does not mean that the use of the Hencky strain is inappropriate. The rate expressed on the X_i

axes depends on the choice of the rotating axes on which the strain is observed. Equation (12) is the result when the observation axes are the principal axes x_i^L characterized by the orthogonal rotation tensor $\mathbf{T}$. If we observe the same deformation on a frame composed of rotating axes different from x_i^L , the rate described on the X_i axes becomes different from $\dot{\mathbf{h}}_L$ [8,9]. In addition to the frame coincident with x_i^L , there are an infinite number of frames [14]. If we choose an appropriate rotating frame to observe the simple-shear deformation, we may obtain $\mathbf{D}$ given by Equation (16) as the strain rate of the simple-shear deformation. The rotating frame satisfying this condition, denoted as the $\mathbf{D}$ -frame [8], was actually found in the mid-1990s [8,14]. This means that it has been proved that the rate of deformation $\mathbf{D}$ is the objective rate and the strain rate of the Hencky strain [8,14]. Now it is understood that only the Hencky strain observed on the $\mathbf{D}$ -frame gives reliable information to discuss the kinematics of large deformations [14]. The problem of $\dot{\mathbf{h}}_L$ is not a problem arising from the use of the Hencky formalism, but from the choice of the appropriate rotating frame. For details of the $\mathbf{D}$ -frame and recent developments in the area of Hencky strains, see [8,9,14–16].

At present, the Hencky strain is a favored measure of strain because of its remarkable properties at large deformations. Many authors have recently used the Hencky strain in constitutive modeling of solids. A list of such studies is found, for example, in the references of [17]. Shrivastava et al. [4] have recently stated that the Hencky formulation does not apply to large simple-shear deformation. This claim is contrary to recently obtained knowledge about the Hencky strain. As they have cited a paper published in 1937 [18] to support their claim, it is clear that they neglect recent developments in the Hencky strain.

6. Hencky equivalent strain

For a plastic strain $\boldsymbol{\varepsilon} = (\varepsilon_{ij})$ satisfying $\varepsilon_{ii} = 0$, its equivalent strain ε_{eq} is given by

$$\varepsilon_{\text{eq}} = \sqrt{\frac{2}{3} \varepsilon_{ij} \varepsilon_{ij}} = \frac{\sqrt{2}}{3} \sqrt{(\varepsilon_{\text{I}} - \varepsilon_{\text{II}})^2 + (\varepsilon_{\text{II}} - \varepsilon_{\text{III}})^2 + (\varepsilon_{\text{III}} - \varepsilon_{\text{I}})^2}, \quad (20)$$

where ε_{I} , ε_{II} and ε_{III} are the principal strains of $\boldsymbol{\varepsilon}$. From Equations (4), (5), (7) and (20), we have the same Hencky equivalent strain written as

$$h_{\text{eq}} = \frac{2}{\sqrt{3}} \ln \left(\frac{\gamma}{2} + \sqrt{1 + \frac{\gamma^2}{4}} \right) \quad (21)$$

for both $\mathbf{h}_L$ and $\mathbf{h}_E$. Differentiating Equation (21) with respect to γ , the Hencky equivalent strain increment dh_{eq} is given by

$$dh_{\text{eq}} = \frac{2}{\sqrt{3}} \frac{d\gamma}{\sqrt{4 + \gamma^2}}. \quad (22)$$

On the other hand, Shrivastava et al. have derived their Hencky equivalent strain increment dh_{eq}^S from $d\mathbf{h}_L$ of Equation (11), which is written as

$$dh_{\text{eq}}^S = \frac{d\gamma}{\sqrt{3}} \sqrt{\frac{4}{4 + \gamma^2} + \frac{16}{(4 + \gamma^2)^2} \left[\ln \left(\frac{\gamma}{2} + \sqrt{1 + \frac{\gamma^2}{4}} \right) \right]^2}. \quad (23)$$

They have found that

$$dh_{eq}^S \neq dh_{eq} \quad \text{and} \quad \int_0^\gamma dh_{eq}^S \neq h_{eq}(\gamma). \quad (24)$$

The discrepancy between dh_{eq}^S and h_{eq} may be a reason why they have concluded that the Onaka paper in which Equation (21) has been derived is incorrect. However, in addition to the fact that $\dot{\mathbf{h}}_L$ shown by Equation (12) is not actually the strain rate, dh_{eq}^S derived from $d\mathbf{h}_L$ is not the Hencky equivalent strain increment.

The same equivalent strain is obtained from both $\mathbf{h}_L$ and $\mathbf{h}_E$. From Equations (5), (7) and (16), $\mathbf{h}_E(\gamma + d\gamma)$ is given by

$$\mathbf{h}_E(\gamma + d\gamma) = f(\gamma) \begin{pmatrix} \gamma & 2 & 0 \\ 2 & -\gamma & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & d\gamma/2 & 0 \\ d\gamma/2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (25)$$

The Hencky equivalent strain $h_{eq}(\gamma + d\gamma)$ given from Equation (20) is written as

$$h_{eq}(\gamma + d\gamma) = \frac{2}{\sqrt{3}} f(\gamma) \sqrt{\gamma^2 + \left(2 + \frac{d\gamma}{2f(\gamma)}\right)^2} \approx \frac{2}{\sqrt{3}} \ln \left(\frac{\gamma}{2} + \sqrt{1 + \frac{\gamma^2}{4}} \right) + \frac{2}{\sqrt{3}} \frac{d\gamma}{\sqrt{4 + \gamma^2}}. \quad (26)$$

Hence we have

$$dh_{eq} = h_{eq}(\gamma + d\gamma) - h_{eq}(\gamma) = \frac{2}{\sqrt{3}} \frac{d\gamma}{\sqrt{4 + \gamma^2}}, \quad (27)$$

which is identical to Equation (22). This result could also be obtained from Equation (10) [3]. In [4], Shrivastava et al. have claimed that the Hencky equivalent strain shown in [3] is derived neglecting the rotation of the principal axes during simple-shear deformation. As shown by Equations (25) and (26), the increment dh_{eq} given by Equation (27) is that explained by the concept of the $\mathbf{D}$ -frame. The claim by Shrivastava et al. about dh_{eq} is not true.

Shrivastava et al. [4], Jonas et al. [19] and their research group [20,21] have recommended the nominal equivalent strain given by

$$e_{eq} = \gamma/\sqrt{3} \quad (28)$$

to characterize large simple-shear deformation. Considering well-known major problems with the nominal strain when we treat large deformations, there are no rational reasons to use the nominal equivalent strain instead of the Hencky equivalent strain. As Onaka has shown, the Hencky equivalent strain is the appropriate quantity to represent the degree of large simple-shear deformation [2,3].

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References

- [1] B.S. Altan (ed.), *Severe Plastic Deformation: Toward Bulk Production of Nanostructured Materials*, Nova Science Publishers, New York, 2006.
- [2] S. Onaka, J. Jpn. Inst. Met. 74 (2010) p.165.
- [3] S. Onaka, Phil. Mag. Lett. 90 (2010) p.633.
- [4] S. Shrivastava, C. Ghosh and J.J. Jonas, Phil. Mag. 92 (2012) p.779.
- [5] R.J. Asaro and V.A. Lubarda, *Mechanics of Solids and Materials*, Cambridge University Press, New York, 2006, p.55.
- [6] A.D. Freed, J. Eng. Mater. Tech. 117 (1995) p.379.
- [7] P. Lancaster and M. Tismenetsky, *The Theory of Matrices: With Applications*, 2nd ed., Academic Press, San Diego, 1985, p.304.
- [8] W.D. Reinhardt and R.N. Dubey, Mech. Res. Comm. 22 (1995) p.165.
- [9] W.D. Reinhardt and R.N. Dubey, Trans. ASME 118 (1996) p.692.
- [10] J.E. Fitzgerald, J. Appl. Phys. 51 (1980) p.5111.
- [11] M.E. Gurtin and K. Spear, Int. J. Solid Struct. 19 (1983) p.437.
- [12] A. Hoger, Int. J. Solid Struct. 22 (1986) p.1019.
- [13] A. Hoger, Int. J. Solid Struct. 23 (1987) p.1645.
- [14] H. Xiao, O.T. Bruhns and A. Meyers, Acta Mech. 124 (1997) p.89.
- [15] H. Xiao, O.T. Bruhns and A. Meyers, Int. J. Solid Struct. 35 (1998) p.4001.
- [16] H. Xiao, O.T. Bruhns and A. Meyers, Mech. Res. Comm. 25 (1998) p.59.
- [17] J. Arghavani, F. Auricchio and R. Naghdabadi, Int. J. Plast. 27 (2011) p.940.
- [18] A. Nadai, J. Appl. Phys. 8 (1937) p.205.
- [19] J.J. Jonas, C. Ghosh and S.C. Shrivastava, Mater. Trans. 52 (2011) p.1748.
- [20] S.C. Shrivastava, J.J. Jonas and G. Canova, J. Mech. Phys. Solid 30 (1982) p.75.
- [21] J.J. Jonas, C. Ghosh, V. Basabe and S. Shrivastava, Phil. Mag., in press (2012), DOI:10.1080/14786435.2012.671547.